Development of a Cusped Leading-Edge Airfoil CFD Case in OpenFOAM 13

Project progress report and reproducibility record

Status date: 28 October 2026 **Case directory:** ~/Projects/cusped-airfoil/openfoam-case **Solver:** OpenFOAM Foundation 13, foamRun with shockFluid

Current milestone: The custom airfoil geometry has been checked, a structured multi-block 2D mesh has been generated, the mesh passes the default checkMesh checks, and a short transient shockFluid startup run completed to 2e-05 s. The case has not yet been validated against the paper or used to report final aerodynamic coefficients.

Executive summary

The goal is to build an OpenFOAM case for the cusped-leading-edge airfoil described by Akram and Hasan (2024). Rather than building every configuration file from scratch, the workflow reuses two tutorials shipped with the installed OpenFOAM 13 distribution:

- fluid/aerofoilNACA0012 as a reference for a structured multi-block airfoil mesh and the blockMesh → transformPoints → extrudeMesh sequence.
- shockFluid/diffuserIntake as a reference for the shockFluid solver configuration, thermophysical properties, and numerical settings.

The NACA tutorial's grid was adapted to project onto the custom cusped airfoil, while the diffuser example's flow setup was adapted to a laminar ideal-gas case. The initial mesh showed excessive skewness near the sharp leading edge. Refining the upstream block reduced the maximum skewness from 8.715 to 3.082, and the final checkMesh output reported Mesh OK. However, maximum non-orthogonality remains high at about 87.27°, with 186 faces above 70°. That is a mesh-quality concern to revisit before drawing quantitative conclusions.

The first solver run completed 200 time steps from 0 to 2e-05 s, writing fields at 5e-06, 1e-05, 1.5e-05, and 2e-05 s. This demonstrates that the case can initialise and advance; it does not establish physical convergence, grid independence, or agreement with the paper.

1. Project objective and paper reference

The assignment is to create the airfoil geometry reported in the paper and run at least a basic OpenFOAM simulation around it. The longer-term objective is to investigate the paper's supersonic flow case.

The reference paper studies unsteady, viscous, laminar compressible flow around a new symmetric cusped-leading-edge airfoil. Parameters relevant to this case include:

- Chord length: c = 1 m.
- Reynolds number: nominally Re = 5 × 10^5.
- Mach number range: 1.25 to 2.13.
- Angles of attack discussed: 0°, 10°, and 20°.
- Maximum thickness is approximately 0.05c, with the cited maximum-thickness location at x/c = 0.75.
- The paper uses an O-grid and a far-field diameter of 80 chord lengths in its final simulations.

The present OpenFOAM case is a learning and startup case adapted from tutorials; it is not yet an exact reproduction of the paper's domain, grid, boundary conditions, or numerical methods.

2. Software and reference cases

- **OpenFOAM version:** Foundation OpenFOAM 13 (13-cde978a97c93).
- **Custom geometry source:** cusped_airfoil.stl; a coordinate file cusped_le_akram_hasan.dat is also present one directory above the case.
- **Mesh reference:** $FOAM_TUTORIALS/fluid/aerofoilNACA0012.
- **Solver reference:** $FOAM_TUTORIALS/shockFluid/diffuserIntake.
- **Selected solver:** foamRun using shockFluid, a density-based compressible-flow solver module.

The reference cases were used as templates, not copied as if they represented the same geometry or physical model. In particular, diffuserIntake uses a turbulence model, whereas the paper describes laminar flow. The local case was changed to simulationType laminar.

3. Geometry preparation

3.1 Original STL check

The original geometry was placed at:

/home/sankara/Projects/cusped-airfoil/cusped_airfoil.stl

It was copied into the case under constant/triSurface/cusped_airfoil.stl. The surfaceCheck output reported:

- 2,388 triangles and 1,196 vertices.
- Bounding box: (0 -0.0489542 0) to (1 0.0489542 0.1) metres.
- No illegal triangles.
- Closed surface, with every edge connected to two faces.
- One connected part and one region.

This indicated that the STL was closed and suitable for further preparation, although surfaceCheck also reported very small triangles/close point pairs near the sharp geometry.

3.2 Coordinate-system transformation

The original STL profile lay in the XY plane and was extruded along Z. The airfoil tutorial expects the profile in XZ, with the thin spanwise direction along Y. A transformed copy was created using the equivalent of:

```
surfaceTransformPoints "Rx=90, translate=(0 0.05 0)" \
    constant/triSurface/cusped_airfoil.stl \
    constant/triSurface/cusped_airfoil_xz.stl
```

The transformed surface passed surfaceCheck. Its bounding box was:

(0 -0.05 -0.0489542) to (1 0.05 0.0489542).

The geometry was then scaled in the Y direction by a factor of ten to create a spanwise copy for the tutorial's projection construction:

```
surfaceTransformPoints "scale=(1 10 1)" \
    constant/triSurface/cusped_airfoil_xz.stl \
    constant/geometry/cusped_airfoil_span1.stl
```

The resulting bounding box was:

(0 -0.5 -0.0489542) to (1 0.5 0.0489542).

The geometry check again reported a closed, connected surface without illegal triangles.

4. Mesh generation and quality improvement

4.1 Mesh workflow

The current system/blockMeshDict was copied from the OpenFOAM 13 fluid/aerofoilNACA0012 tutorial and adapted to the custom STL. The filename was changed to cusped_airfoil_span1.stl, and the upper/lower reference points were set to x = 0.75, z ≈ ±0.0489542. The mesh uses six hexahedral blocks with projected edges and a curved upstream boundary.

The workflow was:

```
blockMesh
transformPoints "scale=(1 0 1)"
extrudeMesh
checkMesh
```

In simple terms:

- blockMesh generates the structured block grid around the airfoil profile.
- transformPoints collapses the spanwise coordinate to form the 2D construction plane.
- extrudeMesh creates a thin one-cell-thick mesh for the 2D calculation.
- checkMesh checks topology, cell volumes, non-orthogonality, skewness, and other mesh metrics.

The final mesh has front and back patches of type empty, and the mesh reports two active directions (1 0 1), which is consistent with a 2D XZ-plane model.

4.2 Mesh controls used

The relevant values in system/blockMeshDict at the latest successful mesh stage were:

Setting	Value	Purpose
xMin, xMax	-4, 4	Intended horizontal domain controls in the dictionary
zMin, zMax	-2, 2	Vertical extent controls
zCells	80	Cells between the airfoil region and outer boundary
xUCells	80	Cells in the upstream block; increased from 30 via 60 to 80
xMCells	30	Middle-block cells
xDCells	40	Downstream-block cells
zGrading	40	Vertical cell grading
xUGrading	2	Upstream-block grading; changed from 5 to 2
leadGrading	0.2	Grading control toward the leading edge
xDGrading	10	Downstream grading

Important observed detail: the final checkMesh bounding box was (-1.25 -0.1 -2) to (4 0 2), not (-4 …). The upstream vertices project onto the cylinder, whose centre is at x = 0.75 and radius 2 m; its leftmost point is x = 0.75 - 2 = -1.25 m. The actual domain dimensions should therefore be checked in the dictionary and deliberately revised before a research-quality comparison.

4.3 Mesh-quality iterations

The badly shaped faces were diagnosed from the VTK surfaces written by checkMesh -writeSurfaces -surfaceFormat vtk. No graphical inspection was required to get their coordinates. The skewed and non-orthogonal faces clustered close to the leading edge (x near zero, z near zero), rather than being distributed across the whole mesh.

Mesh stage	Cells	Maximum skewness	Faces flagged for skewness	Maximum non-orthogonality	Faces above 70°	Outcome
Initial tutorial adaptation	16,000	8.71546	12	84.7284°	60	Failed one mesh check
Upstream cells 60; grading 2	20,800	4.2129	1	86.756°	138	Still failed skewness check
Upstream cells 80; grading 2	24,000	3.08223	0	87.2692°	186	Mesh OK reported

The changes were made by editing xUCells and xUGrading. Increasing the upstream resolution reduced skewness sufficiently to pass the default check. However, the final non-orthogonality is very high and the count of faces above 70° increased. The phrase Mesh OK means the mesh passed the active checks in that run; it is not a guarantee that all mesh-quality concerns have been resolved. The mesh should be improved or its suitability investigated before publishing quantitative aerodynamic results.

5. Flow solver and physical model

5.1 Tutorial-based setup

The following files were copied from $FOAM_TUTORIALS/shockFluid/diffuserIntake as a starting point:

- system/controlDict
- system/fvSchemes
- system/fvSolution
- constant/physicalProperties

The original case's controlDict was backed up as system/controlDict.mesh-backup. constant/momentumTransport was written as a laminar model rather than retaining the turbulent kOmegaSST model used by the reference diffuser case.

5.2 Thermophysical properties

The case uses an ideal-gas thermophysical model with Sutherland viscosity and sensible internal energy. The current settings recorded in constant/physicalProperties are:

Property	Setting	Meaning
Thermo type	hePsiThermo	Compressible thermophysical model
Equation of state	perfectGas	Ideal-gas density relation
Transport model	sutherland	Temperature-dependent viscosity relation
Energy model	sensibleInternalEnergy	Internal-energy formulation
Molecular weight	28.966	Air molecular weight in kg/kmol
Cv	719.4	Constant-volume specific heat in J/(kg K)
Sutherland As	1.458e-06	Sutherland transport constant
Sutherland Ts	110.4	Sutherland temperature in K

Cv was changed from the tutorial value of 717 to 719.4, and molecular weight from 28.96 to 28.966. This makes Cp = Cv + R approximately 1006.4 J/(kg K) for air, close to the paper's listed value.

5.3 Initial and boundary fields

The 0 directory contains p, T, and U fields:

- Initial pressure: pInf = 1775.544 Pa.
- Initial temperature: TInf = 300 K.
- Initial freestream velocity: UInf = 433.9 m/s along +x, representing a nominal Mach 1.25 case at zero angle of attack.

Current boundary conditions:

Patch	Pressure p	Temperature T	Velocity U
inlet	Fixed value	Fixed value	Fixed vector (433.9 0 0) m/s
outlet	Zero gradient	Zero gradient	Zero gradient
aerofoil	Zero gradient	Zero gradient (adiabatic treatment)	noSlip
front, back	Constraint types included	Constraint types included	Constraint types included

The field files include #includeEtc "caseDicts/setConstraintTypes", which supplies constraint conditions such as empty for the front/back patches.

5.4 Parameter interpretation that must be documented

The paper reports 1775.544 Pa, 300 K, and velocity components (407.6935, 148.3887) m/s in its example for M = 1.25, Re = 5 × 10^5, and alpha = 20°. The current startup case reused the pressure and temperature, but rotated the velocity to (433.9, 0, 0) m/s for alpha = 0°. The velocity magnitude is approximately the same.

Before treating this as a faithful paper setup, verify the paper's pressure convention. Its text calls the value a gauge pressure, while an ideal-gas thermodynamic model generally requires absolute thermodynamic pressure. The present case is therefore best described as an **initial startup configuration based on the paper's reported freestream values**, with pressure interpretation still to be checked.

6. First transient solver run

The solver was launched from the case directory with:

```
set -o pipefail
foamRun 2>&1 | tee log.foamRun
```

The controlDict settings for this run were:

Setting	Value	Explanation
Solver	shockFluid	Density-based compressible-flow solver module
deltaT	1e-7 s	Time-step size
endTime	2e-5 s	Requested end time
writeInterval	5e-6 s	Field output interval

The run completed and wrote output directories at 5e-06, 1e-05, 1.5e-05, and 2e-05 seconds. The log ended with End and reported approximately 5.45 s execution time.

At the last time step, the log showed:

Logged quantity	Value at t = 2e-5 s
Mean Courant number	0.00195243 (approximately)
Maximum Courant number	0.484867 (approximately)

Logged quantity	Value at t = 2e-5 s
Initial residual for Ux	4.17589e-05
Final linear-solver residual for Ux	5.7505e-12
Initial residual for Uz	3.36528e-03
Final linear-solver residual for Uz	1.89575e-11
Initial residual for e	3.03203e-04
Final linear-solver residual for e	5.11423e-11

The density equation was reported as a diagonal update with zero iterations in that step. The maximum Courant number is below 0.5, which is a reasonable sign for this initial time-step test. Very small final residuals indicate the linear equation systems were solved tightly at the reported time steps; they do **not** by themselves prove that the flow is physically converged.

The total simulated time is 20 microseconds. The chord-flow time is approximately:

$c / U = 1 / 433.9 \approx 2.30\text{e-}3$ s.

Thus, the current run covers less than 1% of one chord-flow time. It is a startup/smoke test, not a statistically or physically settled result.

7. What is complete, and what is not

Completed

1. Located and checked the custom STL geometry.
2. Transformed the geometry into the coordinate convention used by the tutorial mesh.
3. Adapted the official NACA0012 tutorial's structured multi-block mesh to the cusped airfoil surface.
4. Used blockMesh, transformPoints, and extrudeMesh to create the 2D-style mesh.
5. Diagnosed mesh-quality problems near the leading edge from checkMesh VTK output.
6. Improved maximum skewness through targeted upstream-block refinement.
7. Adapted the shockFluid/diffuserIntake numerical/thermophysical setup for laminar ideal-gas flow.
8. Completed a short transient solver run and wrote time-directory outputs.

Not yet completed

1. No formal post-processing of Mach number, pressure, temperature, shock location, or wall pressure has been completed.
2. No lift, drag, or aerodynamic coefficients have been integrated.
3. No grid-independence study or time-step-independence study has been completed.
4. The far-field boundary is much closer than the paper's 80-chord-diameter domain.
5. The high maximum non-orthogonality (87.27°, with 186 faces above 70°) needs investigation before trusting wall forces or shock details.
6. The pressure convention and the exact relation between the setup and $Re = 5 \times 10^5$ need to be confirmed.
7. The OpenFOAM numerical method and boundary conditions still need a deliberate comparison with the paper's Fluent setup.

8. How to document an OpenFOAM project reproducibly

OpenFOAM is file- and log-based. The case dictionaries record the current setup, but they do not automatically produce a complete human-readable narrative of every edit. A good record combines the case files, logs, short notes, and a repeatable script.

What the case folders mean

```
openfoam-case/
├── 0/               # Initial fields and boundary conditions
├── constant/        # Geometry, mesh, material and transport models
│   ├── geometry/
│   ├── physicalProperties
│   ├── momentumTransport
│   └── polyMesh/    # Generated mesh
├── system/          # Mesh, solver, numerical and run-control dictionaries
├── 5e-06/           # Results written at the first requested time
├── 1e-05/           # Later solution fields
├── 1.5e-05/
├── 2e-05/
└── log.foamRun      # Captured solver output
```

Capture logs instead of relying on terminal scrollback

For each important operation, save both standard output and errors. In Bash, set -o pipefail makes the pipeline return a failure if the OpenFOAM command fails even when output is piped to tee.

```
set -o pipefail
blockMesh 2>&1 | tee log.blockMesh
transformPoints "scale=(1 0 1)" 2>&1 | tee log.transformPoints
extrudeMesh 2>&1 | tee log.extrudeMesh
checkMesh 2>&1 | tee log.checkMesh
foamRun 2>&1 | tee log.foamRun
```

When a command is already complete, preserve its existing output in the report and avoid rerunning it unless necessary. Rerunning mesh tools can replace the current mesh.

Suggested record-keeping habits

- Keep a short README.md beside the case that explains the objective, reference tutorials, current mesh/solver settings, known issues, and next planned step.
- Keep a dated notes/ or journal/ entry whenever a material setting changes. Record the old value, new value, reason, and resulting check.
- Keep log.blockMesh, log.checkMesh, and log.foamRun for each important run.
- Use an Allrun script once the workflow is stable. It should execute the chosen mesh-generation sequence and solver with captured logs so the run is repeatable.
- If using Git for the case, commit meaningful milestones such as “custom geometry checked”, “mesh quality improved”, and “startup run completed”. Avoid committing huge generated results unless that is intentional.
- Use foamLog log.foamRun where supported by the installed distribution to extract residual histories from the solver log. This supports plots; it does not replace the written explanation of decisions.

9. Recommended next steps

1. **Archive the current working state.** Keep copies of system/blockMeshDict, system/controlDict, constant/physicalProperties, constant/momentumTransport, 0/p, 0/T, 0/U, and the full log.foamRun.

2. **Inspect mesh quality further.** The mesh passes the current default checks, but the non-orthogonality is very high. Examine the leading-edge block construction and consider a better block layout or local mesh design before extracting forces.
3. **Check domain size and coordinate bounds.** The actual box is x = -1.25 ... 4, z = -2 ... 2, with a thin spanwise extent y = -0.1 ... 0; this is much smaller than the paper's 80-chord-diameter far field.
4. **Confirm the thermodynamic pressure convention and Reynolds number.** The ideal-gas model must use a thermodynamically appropriate absolute pressure.
5. **Then begin post-processing.** Calculate Mach number, plot/sample pressure and temperature, inspect shock locations, and add force integration only after the mesh/boundaries are judged adequate.
6. **Only after that, run validation studies.** Vary mesh resolution and time step, then compare quantities against the paper's data.

10. Short status statement for a supervisor

I started from OpenFOAM Foundation 13's aerofoilNACA0012 tutorial to adapt its structured multi-block mesh to the cusped airfoil STL. I validated the STL with surfaceCheck, transformed the geometry into the tutorial's coordinate convention, generated the mesh using blockMesh, transformPoints, and extrudeMesh, and refined the upstream block after checkMesh identified skewness near the leading edge. The final mesh contains 24,000 hexahedral cells and passes the current default checkMesh checks, although high non-orthogonality remains a known limitation. I then used the shockFluid/diffuserIntake tutorial as the basis for an ideal-gas, Sutherland-transport, laminar setup and completed a short transient run to 2e-05 s. This confirms the case can initialise and advance, but post-processing, mesh/time-step independence, domain expansion, and comparison against the paper are still pending.

References

1. Akram, S. and Hasan, N. (2024). “The dynamics of supersonic flow past a new cusped leading edge airfoil.” *Progress in Computational Fluid Dynamics*, 24(1), pp. 1–28. DOI: 10.1504/PCFD.2022.10052925.
2. OpenFOAM Foundation 13 tutorial: tutorials/fluid/aerofoilNACA0012 (local path: $FOAM_TUTORIALS/fluid/aerofoilNACA0012). Source: https://github.com/OpenFOAM/OpenFOAM-13/tree/master/tutorials/fluid/aerofoilNACA0012
3. OpenFOAM Foundation 13 tutorial: tutorials/shockFluid/diffuserIntake (local path: $FOAM_TUTORIALS/shockFluid/diffuserIntake). Source: https://github.com/OpenFOAM/OpenFOAM-13/tree/master/tutorials/shockFluid/diffuserIntake
4. OpenFOAM user-guide overview of cases, running applications, meshing, and post-processing: https://www.openfoam.com/documentation/user-guide